



NOTES

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Module 2

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Syllabus :

MOSFET

Biasing in MOS amplifier circuits:

Fixing VGS, Fixing VG, Drain to Gate feedback resistor.

Small signal operation and modelling:

The DC bias point, signal current in drain, voltage gain, small signal equivalent circuit models, transconductance, The T equivalent circuit model.

MOSFET Amplifier configuration:

Basic configurations, characterizing amplifiers, CS amplifier with and without source resistance, The Common Gate Amplifier, Source follower.

Biasing in MOS Amplifier Circuits

An essential step in the design of a MOSFET amplifier circuit is the establishment of an appropriate dc operating point for the transistor. This is the step known as biasing or bias design. An appropriate dc operating point or bias point is characterized by a stable and predictable dc drain current I_D and by a dc drain-to-source voltage V_{DS} .

Biasing by Fixing V_{GS}

The most straight forward approach to biasing a MOSFET is to fix its gate-to-source voltage V_{GS} to the value required to provide the desired I_D . This voltage value can be derived from the power-supply voltage V_{DD} through the use of an appropriate voltage divider. Alternatively, it can be derived from another suitable reference voltage that might be available in the system. This is *not* a good approach to biasing a MOSFET. To understand the reason for this statement, recall that I_D is

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$$

and note that the values of the threshold voltage V_t , the oxide-capacitance C_{ox} , and (to a lesser extent) the transistor aspect ratio vary widely among devices of supposedly the same size and type. The spread is also large in integrated circuits, especially among devices fabricated on different wafers and certainly between different batches of wafers. Furthermore, both V_t and μ_n depend on temperature, with the result that if we fix the value of V_{GS} , the drain current I_D becomes very much temperature dependent. As shown in the Fig. 5.51 two i_D - v_{GS} characteristic curves representing extreme values in a batch of MOSFETs of the same type having different current I_D for the fixed value of V_{GS} .

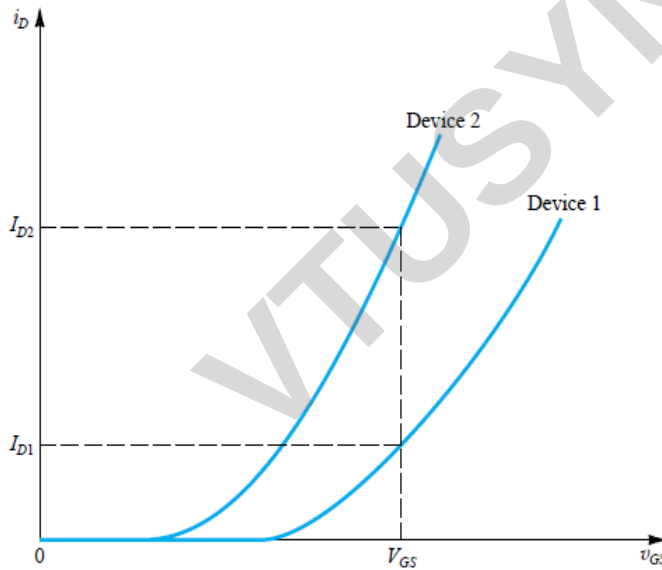


Figure 5.51 The use of fixed bias (constant V_{GS}) can result in a large variability in the value of I_D . Devices 1 and 2 represent extremes among units of the same type.

Biasing by Fixing V_G and Connecting a Resistance in the Source

An excellent biasing technique for discrete MOSFET circuits consists of fixing the dc voltage at the gate, V_G , and connecting a resistance in the source lead, as shown in Fig. 5.52(a). For this circuit we can write

$$V_G = V_{GS} + R_S I_D \quad (5.93)$$

Here the resistor R_S provides *negative feedback*, which acts to stabilize the value of the bias current I_D . To see how this comes about, consider what happens when I_D increases

for whatever reason. Equation (5.93) indicates that since V_G is constant, V_{GS} will have to decrease. This in turn results in a decrease in I_D , a change that is opposite to that initially assumed. Thus the action of R_S works to keep I_D as constant as possible. This negative feedback action of R_S gives it the name **degeneration resistance**.

Figure 5.52(b) provides a graphical illustration of the effectiveness of this biasing scheme. Here too we show the i_D - v_{GS} characteristics for two devices that represent the extremes of a batch of MOSFETs. Observe that compared to the case of fixed V_{GS} , here the variability obtained in I_D is much smaller. Also, note that the variability decreases as V_G and R_S are made larger (thus providing a bias line that is less steep).

Two possible practical discrete implementations of this bias scheme are shown in Fig. 5.52(c) and (e). The circuit in Fig. 5.52(c) utilizes one power-supply V_{DD} and derives V_G through a voltage divider (R_{G1} , R_{G2}). As shown in Fig. 5.52(d) capacitor C_{C1} blocks dc and thus allows us to couple the signal v_{sig} to the amplifier input without disturbing the MOSFET dc bias point. Finally, note that in. When two power supplies are available, as is often the case, the somewhat simpler bias arrangement of Fig. 5.52(e) can be utilized.

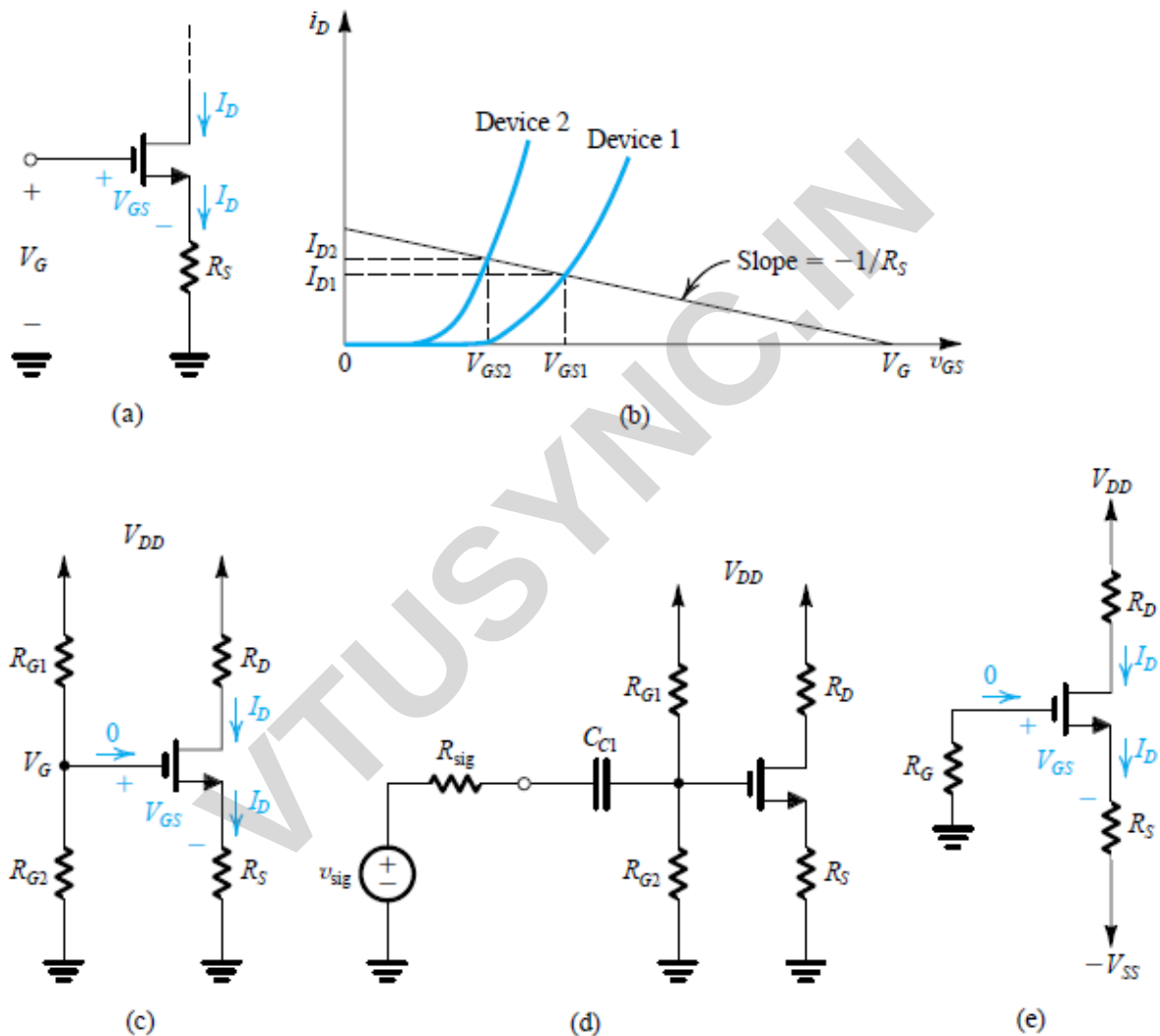


Figure 5.52 Biasing using a fixed voltage at the gate, V_G , and a resistance in the source lead, R_S : (a) basic arrangement; (b) reduced variability in I_D ; (c) practical implementation using a single supply; (d) coupling of a signal source to the gate using a capacitor C_{C1} ; (e) practical implementation using two supplies.

Biasing Using a Drain-to-Gate Feedback Resistor

A simple and effective discrete-circuit biasing arrangement utilizing a feedback resistor connected between the drain and the gate is shown in Fig. 5.54. Here the large feedback

resistance R_G forces the dc voltage at the gate to be equal to that at the drain (because $I_G = 0$). Thus we can write

$$V_{GS} = V_{DS} = V_{DD} - R_D I_D$$

which can be rewritten in the form

$$V_{DD} = V_{GS} + R_D I_D \quad (5.96)$$

which is identical in form to Eq. (5.93), which describes the operation of the bias scheme discussed above [that in Fig. 5.52(a)]. Thus, here too, if I_D for some reason changes, say increases, then Eq. (5.96) indicates that V_{GS} must decrease. The decrease in V_{GS} in turn causes a decrease in I_D , a change that is opposite in direction to the one originally assumed. Thus the negative feedback or degeneration provided by R_G works to keep the value of I_D as constant as possible.

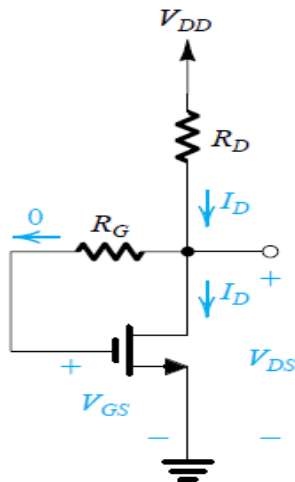


Figure 5.54 Biasing the MOSFET using a large drain-to-gate feedback resistance, R_G .

Small-Signal Operation and Models

For small signal operation we utilize the conceptual amplifier circuit shown in Fig. 5.34. Here the MOS transistor is biased by applying a dc voltage V_{GS} , and the input signal to be amplified, v_{gs} , is superimposed on the dc bias voltage V_{GS} . The output voltage is taken at the drain.

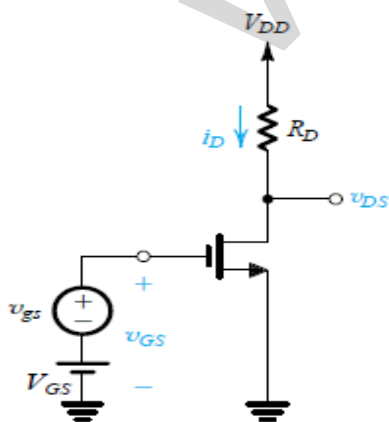


Figure 5.34 Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

The DC Bias Point

The dc bias current I_D can be found by setting the signal v_{gs} to zero; thus,

$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2 = \frac{1}{2} k_n V_{OV}^2 \quad (5.40)$$

Here $V_{OV} = V_{GS} - V_t$ is the overdrive voltage at which the MOSFET is biased to operate. The dc voltage at the drain, V_{DS} , will be

$$V_{DS} = V_{DD} - R_D I_D \quad (5.41)$$

To ensure saturation-region operation, we must have

$$V_{DS} > V_{OV}$$

The Signal Current in the Drain Terminal

Next, consider the situation with the input signal v_{gs} applied. The total instantaneous gate-to-source voltage will be

$$v_{GS} = V_{GS} + v_{gs} \quad (5.42)$$

resulting in a total instantaneous drain current i_D ,

$$\begin{aligned} i_D &= \frac{1}{2} k_n (V_{GS} + v_{gs} - V_t)^2 \\ &= \frac{1}{2} k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t) v_{gs} + \frac{1}{2} k_n v_{gs}^2 \end{aligned} \quad (5.43)$$

The first term on the right-hand side of Eq. (5.43) can be recognized as the dc bias current I_D (Eq. 5.40). The second term represents a current component that is directly proportional to the input signal v_{gs} . The third term is a current component that is proportional to the square of the input signal. This last component is undesirable because it represents *nonlinear distortion*. To reduce the nonlinear distortion introduced by the MOSFET, the input signal should be kept small so that

$$\frac{1}{2} k_n v_{gs}^2 \ll k_n (V_{GS} - V_t) v_{gs}$$

resulting in

$$v_{gs} \ll 2(V_{GS} - V_t) \quad (5.44)$$

or, equivalently,

$$v_{gs} \ll 2V_{OV} \quad (5.45)$$

If this **small-signal condition** is satisfied, we may neglect the last term in Eq. (5.43) and express i_D as

$$i_D \doteq I_D + i_d \quad (5.46)$$

where

$$i_d = k_n (V_{GS} - V_t) v_{gs}$$

The parameter that relates i_d and v_{gs} is the MOSFET transconductance g_m ,

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n (V_{GS} - V_t) \quad (5.47)$$

or in terms of the overdrive voltage V_{OV} ,

$$g_m = k_n V_{OV} \quad (5.48)$$

Figure 5.35 presents a graphical interpretation of the small-signal operation of the MOSFET amplifier. Note that g_m is equal to the slope of the i_D - v_{GS} characteristic at the bias point,

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS} = V_{GS}} \quad (5.49)$$

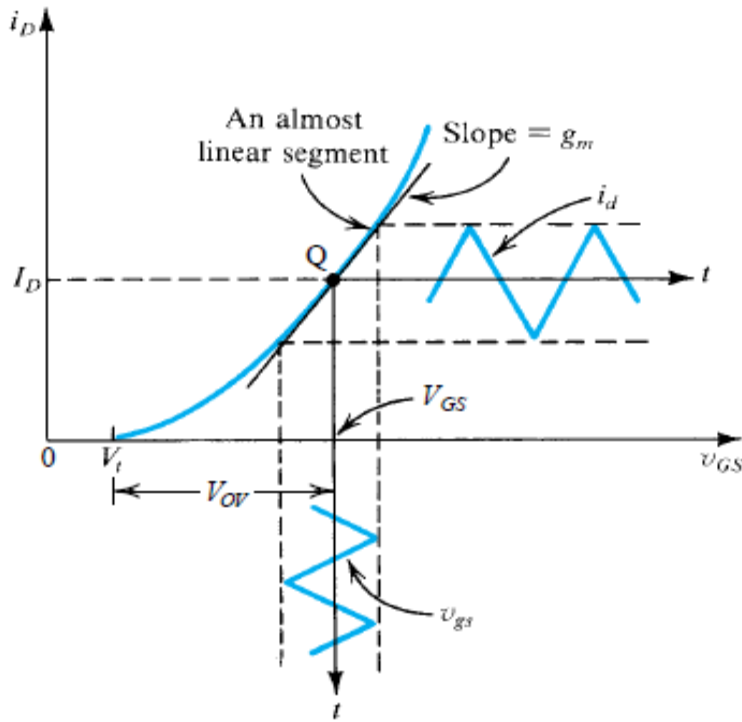


Figure 5.35 Small-signal operation of the MOSFET amplifier.

The Voltage Gain

Returning to the circuit of Fig. 5.34, we can express the total instantaneous drain voltage v_{DS} as follows:

$$v_{DS} = V_{DD} - R_D i_D$$

Under the small-signal condition, we have

$$v_{DS} = V_{DD} - R_D (I_D + i_d)$$

which can be rewritten as

$$v_{DS} = V_{DS} - R_D i_d$$

Thus the signal component of the drain voltage is

$$v_{ds} = -i_d R_D = -g_m v_{gs} R_D \quad (5.50)$$

which indicates that the voltage gain is given by

$$A_v \equiv \frac{v_{ds}}{v_{gs}} = -g_m R_D \quad (5.51)$$

The minus sign in Eq. (5.51) indicates that the output signal v_{ds} is 180° out of phase with respect to the input signal v_{gs} . This is illustrated in Fig. 5.36, which shows v_{GS} and v_{DS} . The input signal is assumed to have a triangular waveform with an amplitude much smaller than $2(V_{GS} - V_t)$, the small-signal condition in Eq. (5.44), to ensure linear operation.

For operation in the saturation region at all times, the minimum value of v_{DS} should not fall below the corresponding value of v_{GS} by more than V_t . Also, the maximum value of v_{DS} should be smaller than V_{DD} ; otherwise the FET will enter the cut off region and the peaks of the output signal waveform will be clipped off.

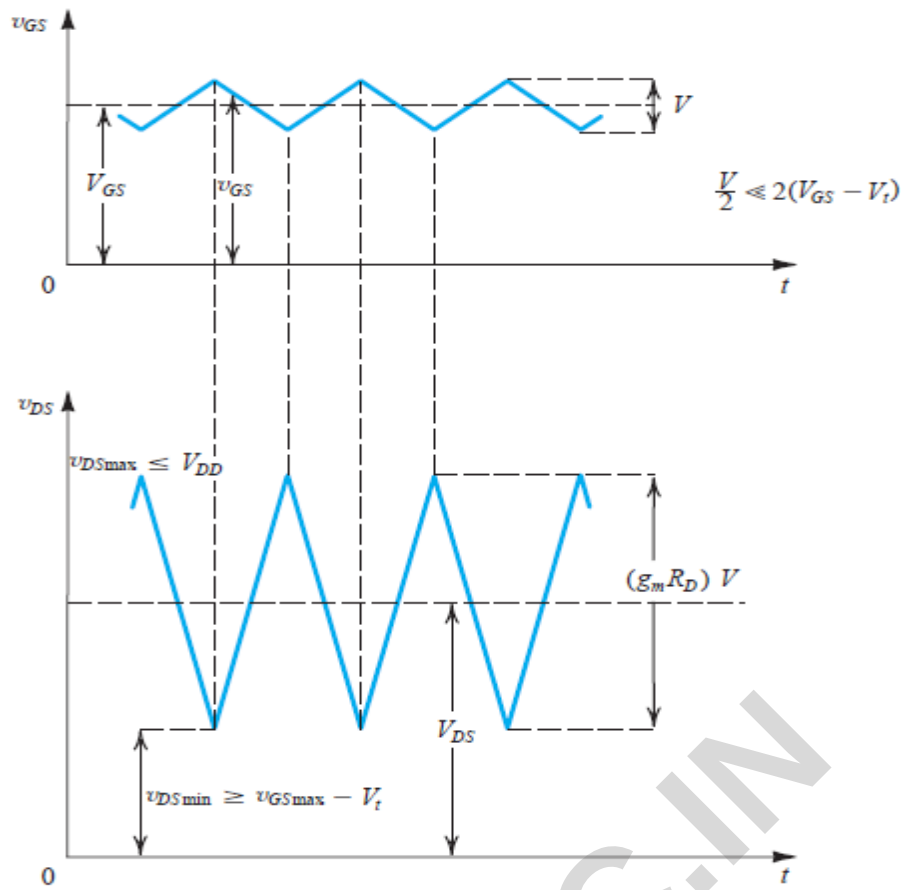


Figure 5.36 Total instantaneous voltages v_{GS} and v_{DS} for the circuit in Fig. 5.34.

Small-Signal Equivalent-Circuit Models

From a signal point of view, the FET behaves as a voltage-controlled current source. It accepts a signal v_{gs} between gate and source and provides a current $g_m v_{gs}$ at the drain terminal. The input resistance of this controlled source is very high—ideally, infinite. The output resistance that is, the resistance looking into the drain also is high, and we have assumed it to be infinite thus far. Putting all of this together, we arrive at the circuit in Fig. 5.37(a), which represents the small-signal operation of the MOSFET and is thus a **small-signal model** or a **small-signal equivalent circuit**.

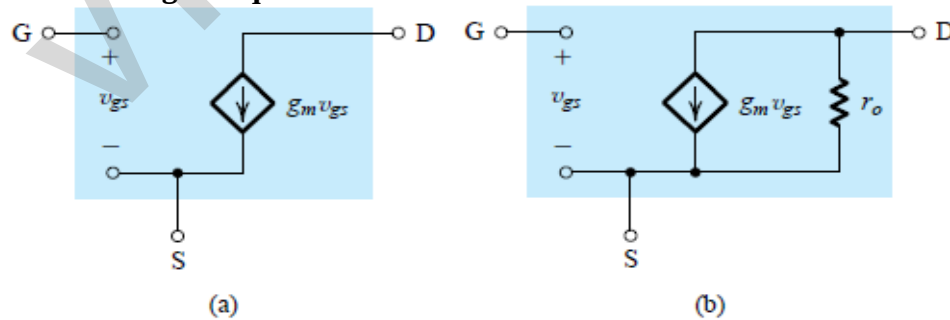


Figure 5.37 Small-signal models for the MOSFET: (a) neglecting the dependence of i_D on v_{DS} in saturation (the channel-length modulation effect); and (b) including the effect of channel-length modulation, modeled by output resistance $r_o = |V_A|/I_D$.

The most serious shortcoming of the small-signal model of Fig. 5.37(a) is that it assumes the drain current in saturation to be independent of the drain voltage. But the drain current does in fact depend on v_{DS} in a linear manner. Such dependence was modeled by a finite resistance r_o between drain and source, whose value was given by

$$r_o = \frac{|V_A|}{I_D} \quad (5.52)$$

The Transconductance g_m

We shall now take a closer look at the MOSFET transconductance given by Eq. (5.47), which we rewrite with $K_n = K_n' (W/L)$ as follows:

$$g_m = k_n' (W/L) (V_{GS} - V_t) = k_n' (W/L) V_{OV} \quad (5.55)$$

This relationship indicates that g_m is proportional to the process g_m parameter and to the W/L ratio of the MOS transistor; hence to obtain relatively large g_m the device must be short and wide.

Another useful expression for g_m can be obtained by substituting for V_{OV} in Eq. (5.55) by $\sqrt{2I_D / (k_n' (W/L))}$ [from Eq. (5.40)]:

$$g_m = \sqrt{2k_n'} \sqrt{W/L} \sqrt{I_D} \quad (5.56)$$

This expression shows two things:

1. For a given MOSFET, g_m is proportional to the square root of the dc bias current.
2. At a given bias current, g_m is proportional to $\sqrt{W/L}$.

Yet another useful expression for g_m of the MOSFET can be obtained by substituting for $k_n' (W/L)$ in Eq. (5.55) by $2I_D / (V_{GS} - V_t)^2$:

$$g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}} \quad (5.57)$$

A convenient graphical construction that clearly illustrates this relationship is shown in Fig. 5.38.

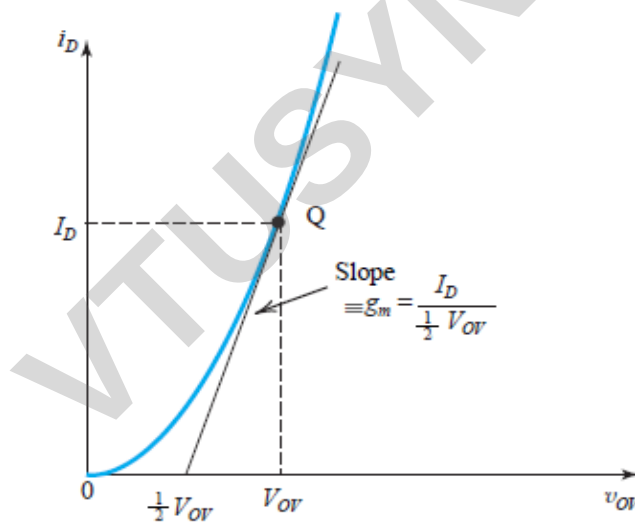


Figure 5.38 The slope of the tangent at the bias point Q intersects the v_{OV} axis at $\frac{1}{2} V_{OV}$. Thus, $g_m = I_D / (\frac{1}{2} V_{OV})$.

The T Equivalent-Circuit Model

Through a simple circuit transformation it is possible to develop an alternative equivalent-circuit model for the MOSFET. The development of such a model, known as the T model, is illustrated in Fig. 5.40. Figure 5.40(a) shows the equivalent circuit studied

above without r_o . In Fig. 5.40(b) we have added a second $g_m v_{gs}$ current source in series with the original controlled source. This addition obviously does not change the terminal currents and is thus allowed. The newly created circuit node, labeled X, is joined to the gate terminal G in Fig. 5.40(c). Observe that the gate current does not change—that is, it remains equal to zero—and thus this connection does not alter the terminal characteristics. We now note that we have a controlled current source $g_m v_{gs}$ connected across its control voltage v_{gs} . We can replace this controlled source by a resistance as long as this resistance draws an equal current as the source. (See the source-absorption theorem in Appendix D.) Thus the value of the resistance is $v_{gs}/g_m v_{gs} = 1/g_m$. This replacement is shown in Fig. 5.40(d), which depicts the alternative model. Observe that i_g is still zero, $i_d = g_m v_{gs}$, and $i_s = v_{gs}/(1/g_m) = g_m v_{gs}$, all the same as in the original model in Fig. 5.40(a).

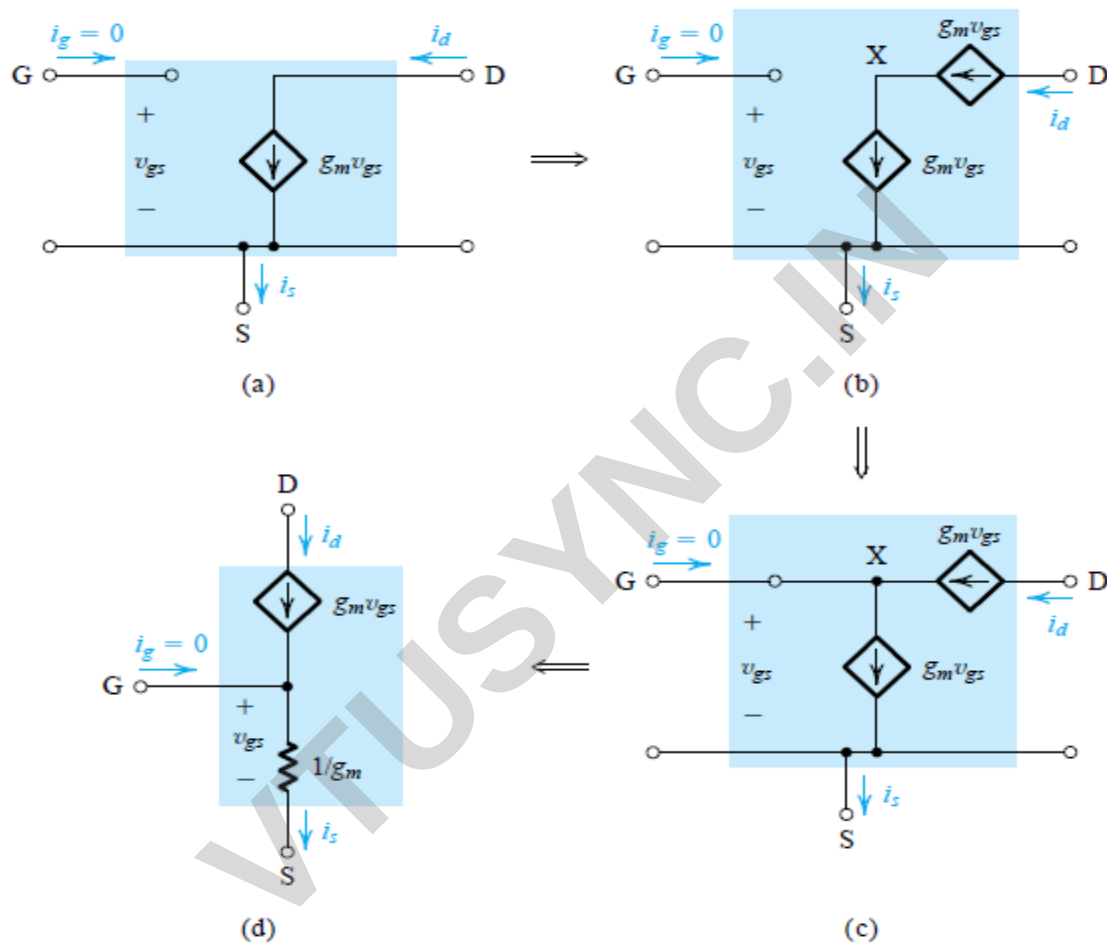


Figure 5.40 Development of the T equivalent-circuit model for the MOSFET. For simplicity, r_o has been omitted; however, it may be added between D and S in the T model of (d).

Basic MOSFET Amplifier Configurations

The Three Basic Configurations:

There are three basic configurations for connecting the MOSFET as an amplifier. Each of these configurations is obtained by connecting one of the three MOSFET terminals to ground, thus creating a two-port network with the grounded terminal being *common* to the input and output ports. Figure 5.43 shows the resulting three configurations with the biasing arrangements omitted.

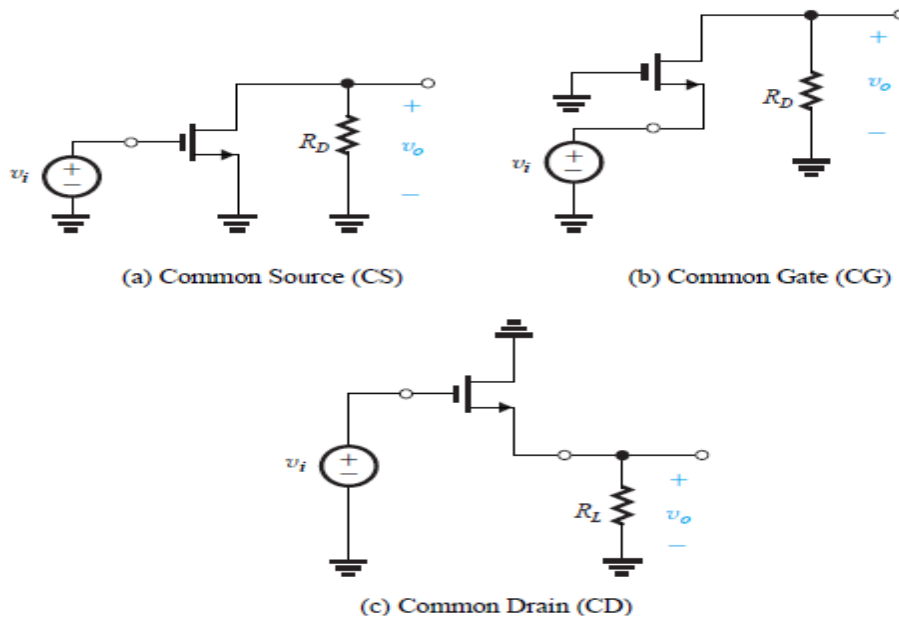


Figure 5.43 The three basic MOSFET amplifier configurations.

1. Common Source Amplifier (CS)

In the circuit of Fig. 5.43(a) the source terminal is connected to ground, the input voltage signal v_i is applied between the gate and ground, and the output voltage signal v_o is taken between the drain and ground, across the resistance R_D . This configuration, therefore, is called the grounded-source or **common-source (CS)** amplifier. It is by far the most popular MOS amplifier

2. Common Gate Amplifier (CG)

The **common-gate (CG)** or grounded-gate amplifier is shown in Fig. 5.43(b). It is obtained by connecting the gate to ground, applying the input v_i between the source and ground, and taking the output v_o across the resistance R_D connected between the drain and ground.

3. Common Drain (CD)

Finally, Fig. 5.43(c) shows the **common-drain (CD)** or grounded-drain amplifier. It is obtained by connecting the drain terminal to ground, applying the input voltage signal v_i between gate and ground, and taking the output voltage signal between the source and ground, across a load resistance R_L . For reasons that will become apparent shortly, this configuration is more commonly called the **source follower**.

Characterizing Amplifiers

Figure 5.44(a) shows an amplifier fed with a signal source having an open-circuit voltage v_{sig} and an internal resistance R_{sig} . These can be the parameters of an actual signal source or, in a cascade amplifier, the Thévenin equivalent of the output circuit of another amplifier stage preceding the one under study. The amplifier is shown with a load resistance R_L

connected to the output terminal. Here, R_L can be an actual load resistance or the input resistance of a succeeding amplifier stage in a cascade amplifier.

Figure 5.44(b) shows the amplifier circuit with the amplifier block replaced by its equivalent-circuit model. The input resistance R_{in} represents the loading effect of the amplifier input on the signal source. It is found from

$$R_{\text{in}} \equiv \frac{v_i}{i_i}$$

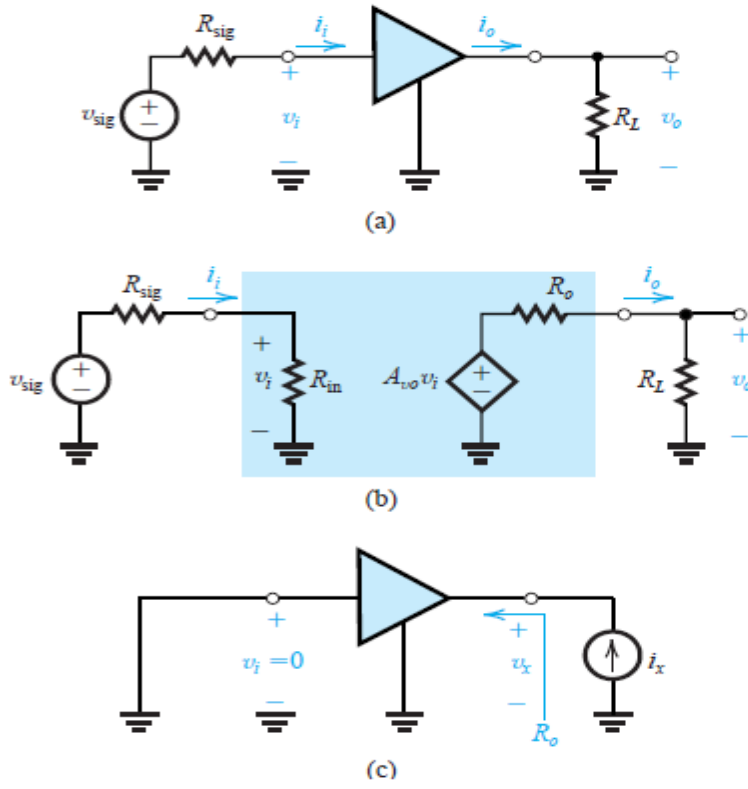


Figure 5.44 Characterization of the amplifier as a functional block: (a) An amplifier fed with a voltage signal v_{sig} having a source resistance R_{sig} , and feeding a load resistance R_L ; (b) Equivalent-circuit representation of the circuit in (a); (c) Determining the amplifier output resistance R_o .

and together with the resistance R_{sig} forms a voltage divider that reduces v_{sig} to the value v_i that appears at the amplifier input,

$$v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig} \quad (5.65)$$

The second parameter in characterizing amplifier performance is the **open-circuit voltage gain** A_{vo} , defined as

$$A_{vo} \equiv \left. \frac{v_o}{v_i} \right|_{R_L = \infty}$$

The third and final parameter is the output resistance R_o . Observe from Fig. 5.44(b) that R_o is the resistance seen looking back into the amplifier output terminal with v_i set to zero. Thus R_o can be determined, at least conceptually, as indicated in Fig. 5.44(c) with

$$R_o = \frac{v_x}{i_x}$$

The controlled source $A_{vo}v_i$ and the output resistance R_o represent the Thévenin equivalent of the amplifier output circuit, and the output voltage v_o can be found from

$$v_o = \frac{R_L}{R_L + R_o} A_{vo}v_i \quad (5.66)$$

Thus the voltage gain of the amplifier proper, A_v , can be found as

$$A_v \equiv \frac{v_o}{v_i} = A_{vo} \frac{R_L}{R_L + R_o} \quad (5.67)$$

and the overall voltage gain G_v ,

$$G_v \equiv \frac{v_o}{v_{sig}}$$

can be determined by combining Eqs. (5.65) and (5.67):

$$G_v = \frac{R_{in}}{R_{in} + R_{sig}} A_{vo} \frac{R_L}{R_L + R_o} \quad (5.68)$$

The Common-Source (CS) Amplifier without source resistance (R_s)

Figure 5.45(a) shows a common-source amplifier (with the biasing arrangement omitted) fed with a signal source v_{sig} having a source resistance R_{sig} . We wish to analyze this circuit to determine R_{in} , A_{vo} , R_o , and G_v . For this purpose we shall assume that R_D is part of the amplifier; thus if a load resistance R_L is connected to the amplifier output, it appears in parallel with R_D .

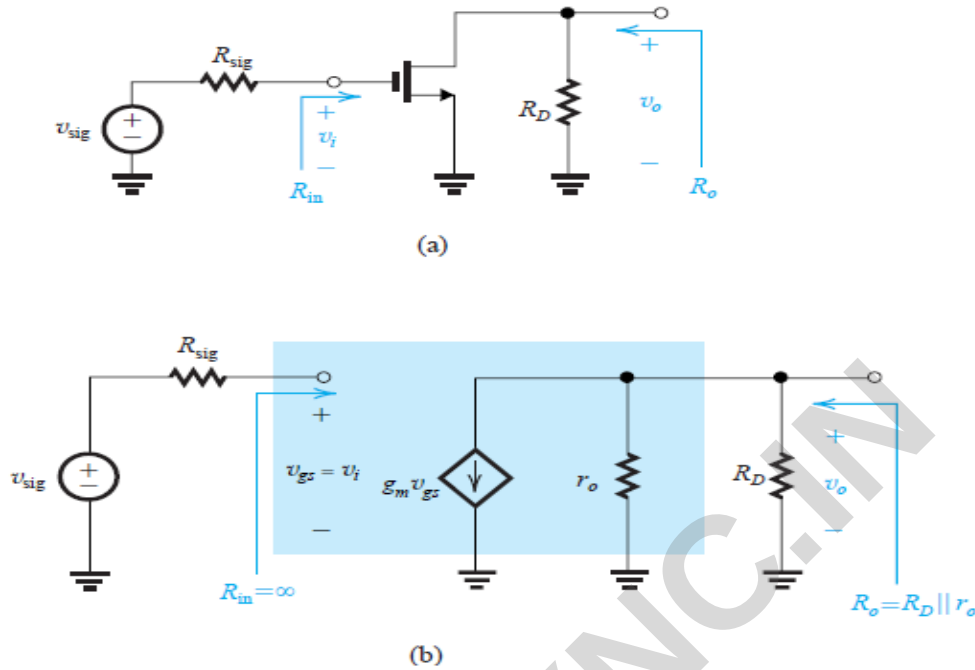


Figure 5.45 (a) Common-source amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig} . The bias circuit is omitted. (b) The common-source amplifier with the MOSFET replaced with its hybrid- π model.

Characteristic Parameters of the CS Amplifier Replacing the MOSFET with its hybrid- π model, we obtain the CS amplifier equivalent circuit shown in Fig 5.45(b). We shall use this equivalent circuit to determine the characteristic parameters R_{in} , A_{vo} , and R_o as follows.

The input resistance R_{in} is obviously infinite,

$$R_{in} = \infty \quad (5.69)$$

The output voltage v_o is found by multiplying the current $(g_m v_{gs})$ by the total resistance between the output node and ground,

$$v_o = -(g_m v_{gs})(R_D \parallel r_o)$$

Since $v_{gs} = v_i$, the open-circuit voltage gain $A_{vo} \equiv v_o/v_i$ can be obtained as

$$A_{vo} = -g_m(R_D \parallel r_o) \quad (5.70)$$

Usually R_D much lower than r_o ($R_D \ll r_o$), thus voltage gain is

$$A_{vo} \approx (-g_m R_D) \quad (5.71)$$

The output resistance R_o is the resistance seen looking back into the output terminal with v_i set to zero. From Fig. 5.45(b) we see that with v_i set to zero, v_{gs} will be zero, and thus $g_m v_{gs}$ will be zero, resulting in

$$R_o = R_D \parallel r_o \quad (5.72)$$

With $R_D \ll r_o$

$$R_o \approx R_D \quad (5.73)$$

Overall Voltage Gain To determine the overall voltage gain G_v , we first note that the infinite input resistance will make the entire signal v_{sig} appear at the amplifier input,

$$v_i = v_{sig} \quad (5.74)$$

Thus overall voltage gain G_v is given by

$$G_v = v_o / v_{sig} = v_o / v_i = A_v$$

It follows that the voltage gain A_v can be obtained by simply replacing R_D in the expression for A_{vo} in Eq. (5.70) by $R_D \parallel R_L$,

$$A_v = -g_m(R_D \parallel R_L \parallel r_o) \quad (5.75)$$

This expression together with the fact that $v_i = v_{sig}$, provides the overall voltage gain,

$$G_v = A_v = -g_m(R_D \parallel R_L \parallel r_o) \quad (5.76)$$

5.27 A CS amplifier utilizes a MOSFET biased at $I_D = 0.25$ mA with $V_{OV} = 0.25$ V and $R_D = 20$ k Ω . The device has $V_A = 50$ V. The amplifier is fed with a source having $R_{sig} = 100$ k Ω , and a 20-k Ω load is connected to the output. Find R_{in} , A_{vo} , R_o , A_v , and G_v . If to maintain reasonable linearity, the peak of the input sine-wave signal is limited to 10% of $(2V_{OV})$ what is the peak of the sine-wave voltage at the output?

Ans. ∞ ; -36.4 V/V; 18.2 k Ω ; -19 V/V; -19 V/V; 0.95 V

The Common-Source (CS) Amplifier with source resistance (R_s)

It is often beneficial to insert a resistance R_s in the source lead of the common-source amplifier as shown in Fig. 5.47(a). The corresponding small-signal equivalent circuit is shown in Fig. 5.47(b), where we note that the MOSFET has been replaced with its T equivalent-circuit model.

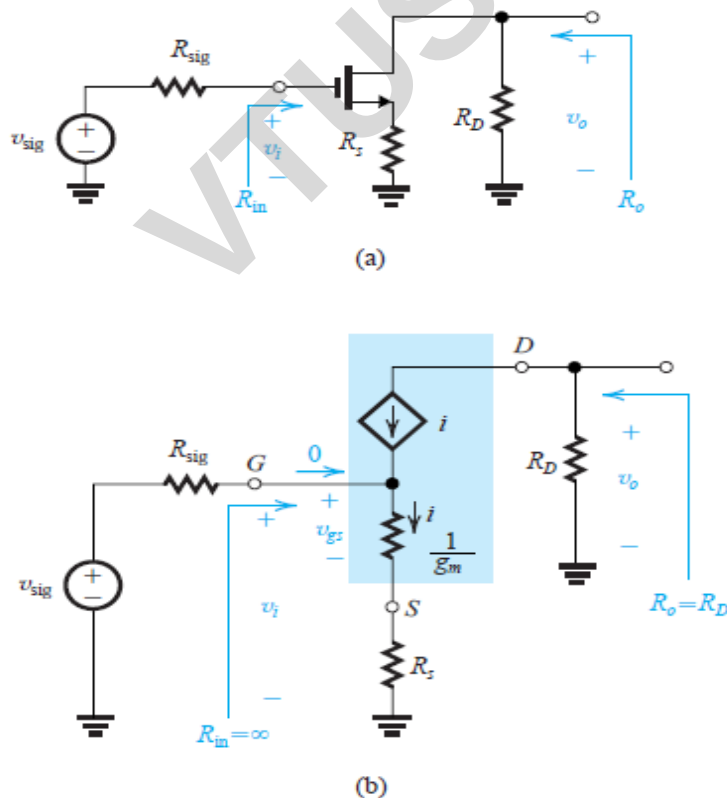


Figure 5.47 The CS amplifier with a source resistance R_s : (a) Circuit without bias details; (b) Equivalent circuit with the MOSFET represented by its T model.

From Fig. 5.47(b) we see that the input resistance R_{in} is infinite and thus $v_i = v_{sig}$. Unlike the CS amplifier, however, here only a fraction of v_i appears between gate and source as v_{gs} . It can be determined from the voltage divider composed of $1/g_m$ and R_s that appears across the amplifier input, as follows:

$$v_{gs} = v_i \frac{1/g_m}{1/g_m + R_s} = \frac{v_i}{1 + g_m R_s} \quad (5.77)$$

The output voltage v_o is obtained by multiplying the controlled-source current i by R_D ,

$$v_o = -i R_D$$

The current i in the source lead can be found by dividing v_i by the total resistance in the source,

$$i = \frac{v_i}{1/g_m + R_s} = \left(\frac{g_m}{1 + g_m R_s} \right) v_i \quad (5.78)$$

Thus, the voltage gain A_{vo} can be found as

$$A_{vo} = \frac{v_o}{v_i} = -\frac{R_D}{1/g_m + R_s} \quad (5.79)$$

which can also be expressed as

$$A_{vo} = -\frac{g_m R_D}{1 + g_m R_s} \quad (5.80)$$

The output resistance R_o which can be found by inspection to be

$$R_o = R_D$$

Alternatively, A_v can be obtained by simply replacing R_D in Eq. (5.79) or (5.80) by $(R_D \parallel R_L)$; thus,

$$A_v = -\frac{R_D \parallel R_L}{1/g_m + R_s} \quad (5.82)$$

or

$$A_v = -\frac{g_m (R_D \parallel R_L)}{1 + g_m R_s} \quad (5.83)$$

The Common-Gate (CG) Amplifier

Figure 5.48(a) shows a common-gate amplifier with the biasing circuit omitted. The amplifier is fed with a signal source characterized by v_{sig} and R_{sig} . Since R_{sig} appears in series with the source, it is more convenient to represent the transistor with the T model than with the π model. Doing this, we obtain the amplifier equivalent circuit shown in Fig. 5.48(b). Note that we have not included r_o : This would have complicated the analysis considerably,

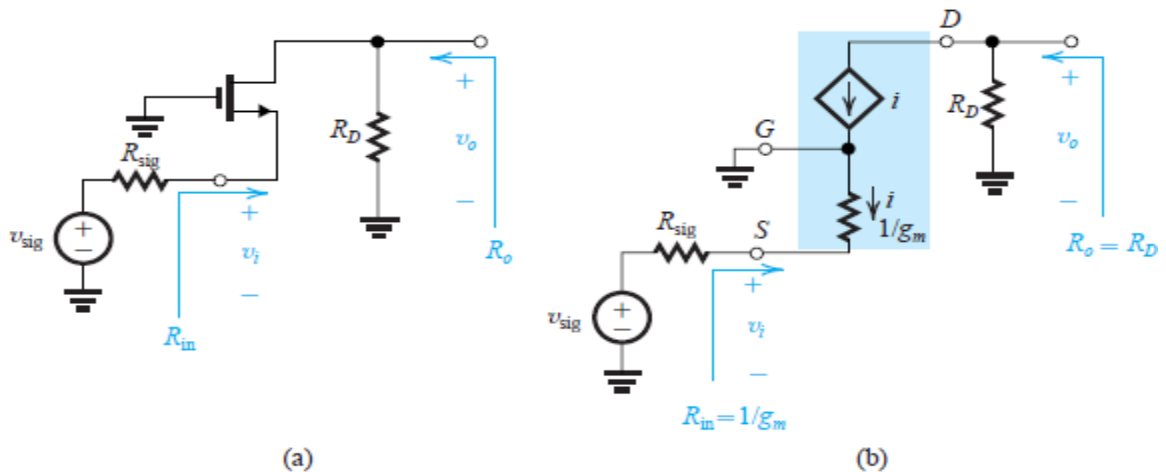


Figure 5.48 (a) Common-gate (CG) amplifier with bias arrangement omitted. (b) Equivalent circuit of the CG amplifier with the MOSFET replaced with its T model.

From inspection of the equivalent circuit of Fig. 5.48(b), we see that the input resistance

$$R_{in} = \frac{1}{g_m} \quad (5.84)$$

To determine the voltage gain A_{vo} , we write at the drain node

$$v_o = -iR_D$$

and substitute for the source current i from

$$i = -\frac{v_i}{1/g_m}$$

to obtain

$$A_{vo} \equiv \frac{v_o}{v_i} = g_m R_D \quad (5.85)$$

The output resistance of the CG circuit can be found by inspection of the circuit in Fig. 5.48(b) as

$$R_o = R_D \quad (5.86)$$

Alternatively, A_v can be obtained by simply replacing R_D in Eq. (5.85) by $(R_D \parallel R_L)$ thus,

$$A_v = g_m (R_D \parallel R_L)$$

Overall voltage gain G_v :

The low input resistance of the CG amplifier can cause the input signal to be severely attenuated. Hence V_{sig} is not equal to v_{in} which is given by

$$\frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = \frac{1/g_m}{1/g_m + R_{sig}} \quad (5.87)$$

Thus overall voltage gain G_v is given by

$$G_v = v_o / v_{sig} = (v_o / v_i)(v_i / v_{sig})$$

$$G_v = \frac{1/g_m}{R_{sig} + 1/g_m} [g_m (R_D \parallel R_L)]$$

Thus,

$$G_v = \frac{(R_D \parallel R_L)}{R_{sig} + 1/g_m} \quad (5.88)$$

The Common-Drain Amplifier or Source Follower

Characteristic Parameters of the Source Follower Figure 5.50(a) shows a source follower with the bias circuit omitted. The source follower is fed with a signal generator (v_{sig}, R_{sig}) and has a load resistance R_L connected between the source terminal and ground.

It is most convenient to use the T model, as shown in Fig. 5.50(b). Note that we have included r_o simply because it is very easy to do so. From the circuit of figure 5.50(a) we can write input resistance as

$$R_{in} = \infty$$

By looking at figure 5.50(c) $v_i = i(1/g_m + R_L)$ and $v_o = i(1/g_m)$ Therefore the A_v is and obtain A_v from the voltage divider formed by $1/g_m$ and R_L as

$$A_v \equiv \frac{v_o}{v_i} = \frac{R_L}{R_L + 1/g_m} \quad (5.89)$$

Setting $R_L = \infty$ we obtain

$$A_{vo} = 1 \quad (5.90)$$

The output resistance R_o is found by setting $v_i = 0$ (i.e., by grounding the gate). Now looking back into the output terminal, excluding R_L , we simply see $1/g_m$, thus

$$R_o = 1/g_m \quad (5.91)$$

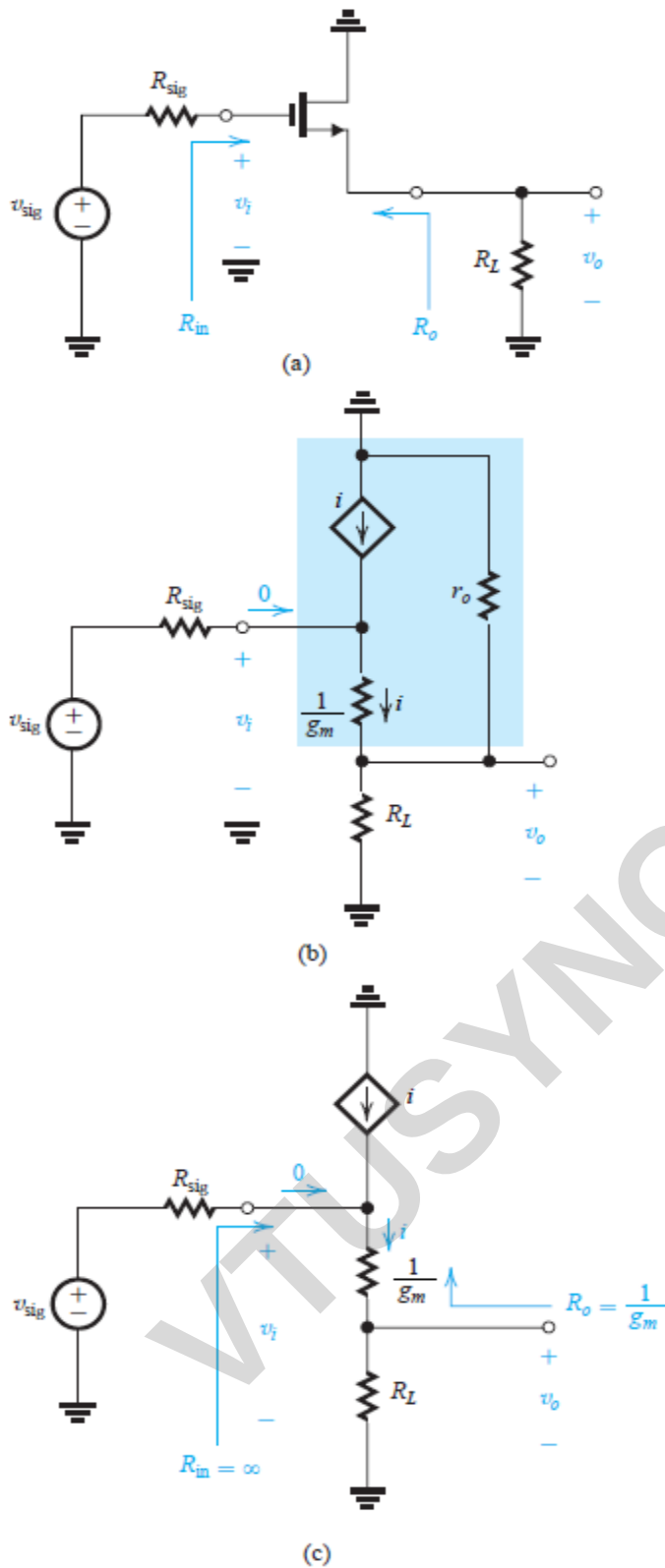


Figure 5.50 (a) Common-drain amplifier or source follower. (b) Equivalent circuit of the source follower obtained by replacing the MOSFET with its T model. Note that r_o appears in parallel with R_L and in discrete circuits, $r_o \gg R_L$. Neglecting r_o , we obtain the simplified equivalent circuit in (c).

The unity open-circuit voltage gain together with R_o in Eq. (5.91) can be used to find A_v when a load resistance R_L is connected. The result is simply the expression in Eq. (5.89). Finally, because of the infinite R_{in} , $v_i = v_{sig}$, and the overall voltage gain is

$$G_v = A_v = \frac{R_L}{R_L + 1/g_m} \quad (5.92)$$

Table 5.4 Characteristics of MOSFET Amplifiers

Amplifier type	Characteristics ^{a, b}				
	R_{in}	A_{vo}	R_o	A_v	G_v
Common source (Fig. 5.45)	∞	$-g_m R_D$	R_D	$-g_m(R_D \parallel R_L)$	$-g_m(R_D \parallel R_L)$
Common source with R_s (Fig. 5.47)	∞	$-\frac{g_m R_D}{1 + g_m R_s}$	R_D	$-\frac{g_m(R_D \parallel R_L)}{1 + g_m R_s}$	$-\frac{g_m(R_D \parallel R_L)}{1 + g_m R_s}$
				$-\frac{R_D \parallel R_L}{1/g_m + R_s}$	$-\frac{R_D \parallel R_L}{1/g_m + R_s}$
Common gate (Fig. 5.48)	$\frac{1}{g_m}$	$g_m R_D$	R_D	$g_m(R_D \parallel R_L)$	$\frac{R_D \parallel R_L}{R_{sig} + 1/g_m}$
Source follower (Fig. 5.50)	∞	1	$\frac{1}{g_m}$	$\frac{R_L}{R_L + 1/g_m}$	$\frac{R_L}{R_L + 1/g_m}$

^a For the interpretation of R_{in} , A_{vo} , and R_o , refer to Fig. 5.44(b).

^b The MOSFET output resistance r_o has been neglected, as is permitted in the discrete-circuit amplifiers studied in this chapter. For IC amplifiers, r_o must always be taken into account.